Md Ahsanul Haque, Ph.D.
MAT-120

Homework-2

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Name Raihan Ahammed

Section 11 ID 2532600043

Instructions:

- Write your full name and section number inside the provided box.
- This assignment contains exactly 5 problems.
- Use correct mathematical notation and clearly show all logical steps.

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1. (10 points) Consider the function:

$$f(x) = \begin{cases} \dfrac{2x^3 - 5x}{x^3 + 1} & \text{if } x < 2 \\ 3 & \text{if } x = 2 \\ \dfrac{x^2 - x - 2}{x - 2} & \text{if } x > 2 \end{cases}$$

   (a) Find $\lim_{x \to 2^-} f(x)$ and $\lim_{x \to 2^+} f(x)$

   (b) Does $\lim_{x \to 2} f(x)$ exist? Justify your answer.

   (c) Find $\lim_{x \to -\infty} f(x)$ and $\lim_{x \to \infty} f(x)$

   (d) Identify all points of discontinuity and classify them.

2. (10 points)

   (a) Let $f(x) = \dfrac{\frac{1}{(x-1)^2} - \frac{1}{4}}{x + 1}$.
   Find $\lim_{x \to -1} f(x)$ using algebraic manipulation.

   (b) Determine if $g(x) = \dfrac{x^3 - 8}{x^2 - 4}$ is continuous at $x = 2$. If not, is the discontinuity removable?

   (c) Find all values of c that make $h(x) = \begin{cases} c^2 x + 2 & \text{if } x \leq 2 \\ cx^2 + 6 & \text{if } x > 2 \end{cases}$ continuous everywhere.

   (d) Investigate the solution of $x^4 - 4x^3 + 2 = 0$ in the interval $[0, 1]$.

3. (10 points) Analyze the following limits:
   (a) $\lim_{x\to 0} \dfrac{1-\cos(5x)}{3x^2}$
   (b) $\lim_{x\to -3} \dfrac{x+3}{\sqrt{x^2+7}-4}$
   (c) $\lim_{x\to -\infty} \frac{\sqrt{25x^2+3x}}{4x-1}$
   (d) $\lim_{x\to 0} x^4 \sin\left(\frac{\pi}{x}\right)$ using the Squeeze Theorem.

4. (10 points) Let $f(x) = x^3 - 6x^2 + 9x$ and $g(x) = \sqrt{3x+4}$
   (a) Find the equation of the tangent line of f at $x = 1$.
   (b) Find the average rate of change of f on $[0, 2]$ and compare it with the instantaneous rate at $x = 1$.
   (c) Use the **limit definition** of the derivative to find $g'(x)$. Show the rationalization step clearly.
   (d) Find the equation of the normal line (the line perpendicular to the tangent line) of g at $x = 4$.

5. (10 points) Consider the curve defined by
$$3x^2 - 2xy + 2y^2 = 14.$$
   (a) Find $\dfrac{dy}{dx}$ using implicit differentiation.
   (b) Find all points where the tangent line is horizontal.
   (c) Find all points where the tangent line is vertical.
   (d) Find $\dfrac{d^2y}{dx^2}$ in terms of x and y.

Solution to the que no. 1

a) When, $x \to 2^-$

$$f(x) = \frac{2x^3 - 5x}{x^3 + 1}$$

$$\therefore \lim_{x \to 2^-} f(x) = \lim_{x \to 2^-} \frac{2x^3 - 5x}{x^3 + 1}$$

$$= \frac{2 \times 2^3 - 5 \times 2}{2^3 + 1}$$

$$= \frac{16 - 10}{9}$$

$$= \frac{2}{3}$$

when, $x \to 2^+$

$$f(x) = \frac{x^2 - x - 2}{x - 2}$$

$$\therefore \lim_{x \to 2^+} f(x) = \lim_{x \to 2^+} \frac{x^2 - x - 2}{x - 2}$$

$$= \frac{2^2 - 2 - 2}{2 - 2} \lim_{x \to 2^+} \frac{x^2 - 2x + x - 2}{x - 2}$$

$$= \lim_{x \to 2^+} \frac{(x - 2)(x + 1)}{(x - 2)}$$

$$= \lim_{x \to 2^+} x + 1$$

$$= 2 + 1$$

$$= 3$$

b) From question 1.a we get,

$$\lim_{x \to 2^-} f(x) = \frac{2}{3}$$

and, $\lim_{x \to 2^+} f(x) = 3$

For, $\lim_{x \to 2} f(x)$,

$$\lim_{x \to 2^-} f(x) \neq \lim_{x \to 2^+} f(x)$$

left hand limit $\neq$ Right hand limit.

∴ Limit does not exist for $\lim_{x \to 2} f(x)$.

c)

$$f(x) = \frac{2x^3 - 5x}{x^3 + 1} \quad ; \quad x < 2$$

$$\therefore \lim_{x \to -\infty} \frac{2x^3 - 5x}{x^3 + 1}$$

$$= \lim_{x \to -\infty} \frac{\frac{2x^3}{x^3} + -\frac{5x}{x^3}}{\frac{x^3}{x^3} + \frac{1}{x^3}}$$

$$= \lim_{x \to -\infty} \frac{2 - \frac{5}{x^2}}{1 + \frac{1}{x^3}}$$

$$= \frac{2 - 0}{1 + 0}$$

$$= 2$$

$$\therefore \lim_{x \to -\infty} f(x) = 2$$

$$f(x) = \frac{x^2-x-2}{x-2} \quad \text{for} \quad x > 2$$

$$\therefore \lim_{x\to\infty} f(x) = \lim_{x\to\infty} \frac{x^2-x-2}{x-2}$$

$$= \lim_{x\to\infty} \frac{(x-2)(x+1)}{(x-2)}$$

$$= \lim_{x\to\infty} (x+1)$$

$$= \infty + 1$$

$$= \infty$$

$$\therefore \lim_{x\to\infty} f(x) = \infty$$

d)

A rational function is discontinuous when denominator is zero.

For, $x < 2$, $f(x) = \frac{2x^3 - 5x}{x^3+1}$

$\Rightarrow x^3 + 1 = 0$

$\Rightarrow x^3 = -1$

$\therefore x = -1$

$\therefore x = -1$ is a discontinuity (Infinite discontinuity).

At, $x=2$, $f(x)=3$

$$\lim_{x\to 2^-} f(x) = \frac{2}{3}$$

and,

$$\lim_{x\to 2^+} f(x) = 3$$

$\therefore$ $x=2$ is Jump discontinuity.

For, $x>2$ $f(x) = \dfrac{x^2-x-2}{x-2}$

$x-2=0$

$\therefore x=2$

$\therefore$ for $x=2$ we get jump discontinuity.

Solution to the question no. -2

a) $$f(x) = \frac{\frac{1}{(x-1)^2} - \frac{1}{4}}{x+1}$$

$$\lim_{x \to -1} f(x) = \lim_{x \to -1} \frac{\frac{1}{(x-1)^2} - \frac{1}{4}}{x+1}$$

$$= \lim_{x \to -1} \frac{\frac{4-(x-1)^2}{4(x-1)^2}}{(x+1)}$$

$$= \lim_{x \to -1} \frac{(2+x-1)(2-x+1)}{4(x-1)^2(x+1)}$$

$$= \lim_{x \to -1} \frac{(x+1)(3-x)}{4(x-1)^2(x+1)}$$

$$= \lim_{x \to -1} \frac{(3-x)}{4(x-1)^2}$$

$$= \frac{3-(-1)}{4((-1)-1)^2}$$

$$= \frac{4}{4 \times (-2)^2}$$

$$= \frac{1}{4}$$

$$\therefore \lim_{x \to -1} f(x) = \frac{1}{4}$$

b) Given that,

$$g(x) = \frac{x^3-8}{x^2-4}$$

The denominator is x^2-4

for $x=2$, $x^2-4 = 2^2-4 = 0$

∴ For $x=2$ the denominator is zero.

∴ $g(x)$ is not continuous at $x=2$.

Now,

$$g(2) = \frac{2^3-8}{2^2-4}$$

$$= \frac{0}{0}$$

= undefined.

$$\lim_{x \to 2^-} \frac{x^3-8}{x^2-4} = '+'$$

$$\lim_{x \to 2^+} \frac{x^3-8}{x^2-4} = '+'$$

∴ Left and right limit exist but $g(x)$ is not defined for $x=2$.

∴ At $x=2$, $g(x)$ has removable discontinuity.

c) If $h(x)$ is continuous then,

$$\lim_{x \to 2^-} h(x) = \lim_{x \to 2^+} h(x)$$

$$\Rightarrow \lim_{x \to 2^-} c^2x + 2 = \lim_{x \to 2^+} cx^2 + 6$$

$$\Rightarrow 2c^2 + 2 = 4c + 6$$

$$\Rightarrow 2c^2 - 4c - 4 = 0$$

$$\Rightarrow c^2 - 2c - 2 = 0$$

$$\therefore c = \frac{-(-2) \pm \sqrt{(-2)^2 - 4 \cdot 1 \cdot (-2)}}{2}$$

$$= \frac{2 \pm \sqrt{4 + 8}}{2}$$

$$= \frac{2 \pm 2\sqrt{3}}{2}$$

$$= 1 + \sqrt{3} \text{ and } 1 - \sqrt{3}$$

$\therefore$ For $c = 1 \pm \sqrt{3}$ the function $h(x)$ is continuous.

d) $f(x) = x^4 - 4x^3 + 2 = 0$

For the interval $[0, 1]$

$f(0) = 0 - 4 \times 0 + 2 = 2$

$$\lim_{x \to 0} \quad x^4 - 4x^3 + 2 = 0 - 0 + 2$$

$$= 2 \quad (\text{'+'})$$

$f(1) = 1 - 4 + 2$

$= -1$

$$\lim_{x \to 1} \quad x^4 - 4x^3 + 2 = 1^4 - 4 \times 1^3 + 2$$

$$= 3 - 4$$

$$= -1 \quad (\text{'-'})$$

$\therefore f(x)$ is continuous on $[0, 1]$

$\therefore$ According to Intermediate Value Theorem we can say solution exist for $x^4 - 4x^3 + 2 = 0$ in the interval $[0, 1)$

Solution to the question no.-3

a) $$\lim_{x\to 0} \frac{1-\cos(5x)}{3x^2}$$

$$= \lim_{x\to 0} \frac{2\sin^2\left(\frac{5x}{2}\right)}{\left(\frac{5x}{2}\right)^2} \cdot \frac{1}{3} \cdot \frac{25}{4}$$

$$= \frac{25}{6} \lim_{x\to 0} \frac{\sin^2\left(\frac{5x}{2}\right)}{\left(\frac{5x}{2}\right)^2}$$

$$= \frac{25}{6} \lim_{x\to 0} \left(\frac{\sin\left(\frac{5x}{2}\right)}{\frac{5x}{2}}\right)^2$$

$$= \frac{25}{6} \times 1$$

$$= \frac{25}{6}$$

b)

$$\lim_{x\to -3} \frac{x+3}{\sqrt{x^2+7}-4}$$

$$= \lim_{x\to -3} \frac{(x+3)\left(\sqrt{x^2+7}+4\right)}{\left(\sqrt{x^2+7}-4\right)\left(\sqrt{x^2+7}+4\right)}$$

$$= \lim_{x\to -3} \frac{(x+3)\left(\sqrt{x^2+7}+4\right)}{x^2+7-16}$$

$$= \lim_{x\to -3} \frac{(x+3)\left(\sqrt{x^2+7}+4\right)}{x^2-9}$$

$$= \lim_{x\to -3} \frac{(x+3)\left(\sqrt{x^2+7}+4\right)}{(x+3)(x-3)}$$

$$= \lim_{x\to -3} \frac{\sqrt{x^2+7}+4}{x-3}$$

$$= \frac{\sqrt{9+7}+4}{-3-3}$$

$$= \frac{8}{-6}$$

$$= \frac{-4}{3}$$

c)

$$\lim_{x \to -\infty} \frac{\sqrt{25x^2+3x}}{4x-1}$$

$$= \lim_{x \to -\infty} \frac{-x\sqrt{25+\frac{3}{x}}}{4x-1}$$

$$= \lim_{x \to -\infty} \frac{-x\sqrt{25+\frac{3}{x}}}{x\left(4-\frac{1}{x}\right)}$$

$$= \lim_{x \to -\infty} \frac{-\sqrt{25+\frac{3}{x}}}{4-\frac{1}{x}}$$

$$= \frac{-\sqrt{25}}{4}$$

$$= -\frac{5}{4}$$

d) $\lim_{x \to 0} x^4 \sin\left(\frac{\pi}{x}\right)$

$-1 \leq \sin\left(\frac{\pi}{x}\right) \leq 1$

$\Rightarrow -x^4 \leq x^4 \sin\left(\frac{\pi}{x}\right) \leq x^4$

Now,

$\lim_{x \to 0} -x^4 = 0$

and,

$\lim_{x \to 0} x^4 = 0$

$\therefore 0 \leq \lim_{x \to 0} x^4 \sin\left(\frac{\pi}{x}\right) \leq 0$

By Squeeze Theorem we get,

$\lim_{x \to 0} x^4 \sin\left(\frac{\pi}{x}\right) = 0$

Solution to' question no. 4

a) $f(x) = x^3 - 6x^2 + 9x$

$$f'(x) = \frac{d}{dx}(x^3 - 6x^2 + 9x)$$

$$= 3x^2 - 12x + 9$$

At, $x = 1$,

The slope of tangent line, $m = f'(1) = 3\times1 - 12\times1 + 9$

$$= 0$$

$$f(1) = 1^3 - 6\times1 + 9$$

$$= 4$$

$\therefore$ Eq. of tangent line at $x = 1$ is,

$$y - f(1) = m(x-1)$$

$$\Rightarrow y = 4 + 0(x-1)$$

$$\therefore y = 4$$

b) $f(x) = x^3 - 6x^2 + 9x$

for the interval $[0, 2]$

average rate of change: $\frac{f(b) - f(a)}{b - a}$

$$= \frac{f(2) - f(0)}{2 - 0}$$

$$= \frac{2^3 - 6 \times 2^2 + 9 \times 2 - 0}{2}$$

$$= \frac{8 - 24 + 18}{2}$$

$$= \underline{1}$$

To find instantaneous change at $x = 1$,

$$f'(x) = \frac{d}{dx} x^3 - 6x^2 + 9x$$

$$= 3x^2 - 12x + 9$$

$$f'(1) = 3 - 12 + 9$$

$$= 0$$

c) $g(x) = \sqrt{3x+4}$

$$g'(x) = \lim_{h \to 0} \frac{g(x+h) - g(x)}{h}$$

$$= \lim_{h \to 0} \frac{\sqrt{3(x+h)+4} - \sqrt{3x+4}}{h}$$

$$= \lim_{h \to 0} \frac{(\sqrt{3x+3h+4} - \sqrt{3x+4})(\sqrt{3x+3h+4} + \sqrt{3x+4})}{h(\sqrt{3x+3h+4} + \sqrt{3x+4})}$$

$$= \lim_{h \to 0} \frac{3x+3h+4-3x-4}{h(\sqrt{3x+3h+4} + \sqrt{3x+4})}$$

$$= \lim_{h \to 0} \frac{3h}{h(\sqrt{3x+3h+4} + \sqrt{3x+4})}$$

$$= \lim_{h \to 0} \frac{3}{(\sqrt{3x+3h+4} + \sqrt{3x+4})}$$

$$= \frac{3}{\sqrt{3x+4} + \sqrt{3x+4}}$$

$$= \frac{3}{2\sqrt{3x+4}}$$

$$\therefore g'(x) = \frac{3}{2\sqrt{3x+4}}$$

d) $g(x) = \sqrt{3x+4}$

$g'(x) = \frac{3}{2\sqrt{3x+4}}$ (from question 3.c)

At $x = 4$,

$g(4) = \sqrt{3\times4+4} = 4$

Slope of tangent line, $g'(4) = m_1 = \frac{3}{2\sqrt{3\times4+4}}$

$= \frac{3}{8}$

∴ tangent line,

$y - g(4) = g'(4)(x-4)$

$\Rightarrow y - 4 = \frac{3}{8}(x-4)$

Slope of perpendicular line, m_2.

$\therefore m_1 m_2 = -1$

$\Rightarrow m_2 = \frac{-1}{m_1} = \frac{-1}{\frac{3}{8}} = \frac{-8}{3}$

∴ Equation of normal line at $x = 4$,

$y - 4 = \frac{-8}{3}(x-4)$

$\Rightarrow -3y + 12 = 8x - 32$

$\therefore 8x + 3y - 44 = 0$

Solution to the question no. – 5

a)

$$\frac{d}{dx}(3x^2 - 2xy + 2y^2) = \frac{d}{dx}14$$

$$\Rightarrow \frac{d}{dx}3x^2 - \frac{d}{dx}2xy + \frac{d}{dx}2y^2 = 0$$

$$\Rightarrow 6x - 2x\frac{dy}{dx} - 2y + 4y\frac{dy}{dx} = 0$$

$$\Rightarrow (4y - 2x)\frac{dy}{dx} = 2y - 6x$$

$$\Rightarrow \frac{dy}{dx} = \frac{2y - 6x}{4y - 2x}$$

$$\therefore \frac{dy}{dx} = \frac{y - 3x}{2y - x}$$

b) we get,

$$\frac{dy}{dx} = \frac{y - 3x}{2y - x}$$

If tangent line is horizontal then, numerator = 0

$$\therefore y - 3x = 0$$

$$\therefore y = 3x$$

Now,

$$3x^2 - 2xy + 2y^2 = 14$$

$$\Rightarrow 3x^2 - 2x \cdot 3x + 2(3x)^2 = 14$$

$\Rightarrow 3x^2 - 6x^2 + 18x^2 = 14$

$\Rightarrow x^2 = \frac{14}{15}$

$\therefore x = \pm\sqrt{\frac{14}{15}}$

$\therefore$ The points are: $(x,y) \equiv (x, 3x)$

$= \left(\sqrt{\frac{14}{15}}, 3\sqrt{\frac{14}{15}}\right)$ and $\left(-\sqrt{\frac{14}{15}}, -3\sqrt{\frac{14}{15}}\right)$

c) we get,

$\frac{dy}{dx} = \frac{y-3x}{2y-x}$

if tangent line is vertical then,

$2y - x = 0$

$\therefore 2y = x$

Now,

$3x^2 - 2xy + 2y^2 = 14$

$\Rightarrow 3(2y)^2 - 2 \cdot 2y \cdot y + 2y^2 = 14$

$\Rightarrow 12y^2 - 4y^2 + 2y^2 = 14$

$\Rightarrow y^2 = \frac{14}{10}$

$\therefore y = \pm\sqrt{\frac{7}{5}}$

$\therefore$ points are $(2y, y) \equiv \left(2\sqrt{\frac{7}{5}}, \sqrt{\frac{7}{5}}\right)$ and $\left(-2\sqrt{\frac{7}{5}}, -\sqrt{\frac{7}{5}}\right)$

d) we get,

$$\frac{dy}{dx} = \frac{y-3x}{2y-x}$$

$$\frac{d^2y}{dx^2} = \frac{(2y-x)(y'-3)}{} -$$

$$= \frac{(2y-x)\frac{d}{dx}(y-3x) - (y-3x)\frac{d}{dx}(2y-1)}{(2y-x)^2}$$

$$= \frac{(2y-x)\left(\frac{y-3x}{2y-x} - 3\right) - (y-3x)\left(\frac{2(y-3x)}{2y-x} - 1\right)}{(2y-x)^2}$$

$$= \frac{(2y-x) \times \frac{y-3x-6y+3x}{(2y-x)} - (y-3x)\frac{2y-6x-2y+x}{2y-x}}{(2y-x)^2}$$

$$= \frac{-5y + \frac{5x(y-3x)}{2y-x}}{(2y-x)^2}$$

$$= \frac{\frac{-10y^2+5xy+5xy-15x^2}{2y-x}}{(2y-x)^2}$$

$$= \frac{-15x^2+10xy-10y^2}{(2y-x)^3}$$

$$= \frac{-5(3x^2-2xy+2y^2)}{(2y-x)^3}$$

$$= -\frac{70}{(2y-x)^3} \quad [\because 3x^2-2xy+2y^2 = 14]$$

$$\therefore \frac{d^2y}{dx^2} = \frac{-70}{(2y-x)^3}$$